Annex II. Kyrgyz Republic: Debt Sustainability Analysis Update[1]

The significant depreciation of the som at end-2015 contributed to a further increase in the external debt burden of the Kyrgyz Republic. The public and publicly guaranteed external debt is estimated to slightly exceed 36 percent of GDP and remittances[2] in present value (PV) terms in the next two years, implying that while the Kyrgyz Republic is still at moderate risk of debt distress, it is getting closer to the ultimate threshold of high risk of debt distress. In order to avoid debt being assessed in high risk of debt distress and the ensuing decline in concessional donor funding, the authorities need to remain extremely cautious when contracting and guaranteeing new debt and should resume fiscal consolidation this year without any delays. The authorities broadly concurred with the results of this DSA.

Background

1. The external public debt of the Kyrgyz Republic increased rapidly since 2013, propelled by large scale public investment.[3] Starting from just 44 percent of GDP two years ago, external debt reached 64.6 percent of GDP by end-2015. The acceleration of debt accumulation at the end of last year was mainly driven by the significant depreciation of the som vis-à-vis the U.S. dollar and the weaker than expected nominal GDP growth. At the same time, the delays in PIP disbursements, mainly due to delays in Chinese projects, contained the accumulation of external debt. Looking forward, the external public debt to GDP ratio is expected to rise further to above 70 percent of GDP in the next two years, and gradually decline afterwards dropping below 60 percent of GDP by 2021.

Underlying DSA assumptions

[1] Prepared by the staff of the International Monetary Fund and the International Development Association. Approved by Juha Kähkönen, Masato Miyazaki (IMF), and Satu Kähkönen (IDA).

[2] The Kyrgyz Republic is classified as a large remittances case. In the 2013–15 period remittances averaged 27 percent of GDP and 78 percent of exports of goods and services, well above the 10 percent and 20 percent respective threshold-criteria for the large remittances case.

[3] External public and publicly-guaranteed debt, which includes the debt of 19 largest SOEs, accounted for 97 percent of public debt in 2015.

Text Table AII.1. Kyrgyz Republic: Composition of External Public and Publicly-Guaranteed Debt in 2015

Creditor	In millions of USD	In percent of total external debt	In percent of GDP
Multilaterals	1,532	42.5	27.45
IMF	188	5.2	3.37
IDA	639	17.7	11.45
ADB	576	16.0	10.32
Other Multilateral	128	3.6	2.30
Official Bilaterals	2,057	57.0	36.9
Paris club	739	20.5	13.2
Non-Paris Club	1,318	36.6	23.6
Eximbank of China	1,296	35.9	23.2
Others Non-Paris Club	22	0.6	0.4
Commercial	17	0.5	0.3
TOTAL	3,606	100.0	64.6

Sources: Authorities data and IMF staff calculations.

2. **In 2015, the Kyrgyz economy proved to be more resilient to external economic shocks compared to peer regional countries.** Economic growth remained strong, while the fiscal deficit declined, largely due to one-off revenues and an underperformance in the PIP program. Total loan disbursements (including PIP and support loans) that were supposed to reach about US$360 and US$660 million in 2015 and 2016, respectively, were revised down to US$265 and US$539 million. Despite the weak external environment, the current account deficit narrowed, partly due to the weaker exchange rate and lower oil prices, but also due to reduced import demand under higher EEU tariffs and frontloading of import purchases in 2014.

Box AII.1. Kyrgyz Republic: Macroeconomic Assumptions for 2015–35

GDP growth: Against the backdrop of strong external headwinds, GDP growth is expected to bottom out in 2016, but recover gradually in the following years and each its potential estimated at 5 percent by 2020. Growth in the medium to long run will be supported by public investments in infrastructure and energy and intensified trade relations within the EEU.

Inflation is expected to remain low by historical standards due to subdued energy and food prices and prudent monetary policy and converge to 4 percent in the long run.

The current account deficit dropped to 10.4 percent in 2015, mainly driven by lower oil prices, slightly weaker domestic demand, and the temporary adverse impact of EEU accession on imports (frontloading of imports in 2014). The current account is expected to widen in the near term, in line with the depletion of imported goods inventories and the pick-up in public investment programs. After the unwinding of these programs the current account is expected to stabilize around 10 percent in the medium to long run.

Box AII.1. Kyrgyz Republic: Macroeconomic Assumptions for 2015–35(concluded)

Public investments will be a major source of economic growth in the coming years. PIP disbursements will reach 8.5 percent of GDP in 2016, and gradually decline to around 4 percent in the medium run. Public investments, which predominantly infrastructure and energy projects financed by China, are also expected to increase net FDI to above 8 percent in 2016. In the long run FDI should stabilize at 6 percent of GDP.

The overall fiscal deficit is expected to rise to close to 9 percent this year driven by elevated capital spending. Following the unwinding of large public investment projects the fiscal balance should stabilize gradually at around 1 percent in the long run.

Financing: FDI and concessional borrowing remains the main source of financing for public infrastructure investments. The average grant element of external borrowing is projected to remain above 40 percent in the medium run, but gradually decline to 20 percent by the end of the period. The effective interest rate is expected to remain below 2 percent in the next five years, but converge towards the market rate by the end of the forecast period.

Text Table AII.2. Kyrgyz Republic: Selected Indicators 2015–20

	2015	2016	2017	2018	2019	2020
Real GDP growth						
Current DSA	3.5	3.0	3.0	5.2	4.6	5.7
Previous DSA (First Review)	2.4	3.6	3.1	5.3	5.2	6.0
Overall fiscal balance (percent of GDP)*						
Current DSA	-3.2	-8.8	-5.2	-2.8	-2.0	-0.5
Previous DSA (First Review)	-5.4	-9.1	-4.1	-1.1	0.6	1.3
Current account balance (percent of GDP)						
Current DSA	-10.4	-15.2	-15.3	-12.6	-12.2	-10.1
Previous DSA (First Review)	-16.0	-17.3	-14.9	-14.1	-13.1	-11.4
PIP Disbursements						
Current DSA	249	497	385	310	312	298
Previous DSA (First Review)	344	618	506	431	384	374
Exchange rate (year end)						
Current DSA	75.9	78.9	80.5	82.1	83.8	85.4
Previous DSA (First Review)	71.5	72.9	74.4	75.9	77.4	78.9

*Including onlending

Sources: Authorities data and IMF staff calculations

External DSA

3. The debt outlook is deteriorating and remains vulnerable to large external shocks. The recent depreciation of the national currency has further increased the risks of external debt distress. The ratio of present value of the public and publicly guaranteed (PPG) debt to GDP and remittances is in small breach of the 36 percent threshold in 2016 and 2017, but still within the +/- 5 percent confidence band. While the probability approach indicates that the risk rating changed to high-medium from medium-high in the first review,[4] it shows no breach of the PV debt to GDP and remittances threshold.[5] Thus the overall risk rating of Kyrgyz's external debt remains moderate risk of debt distress. The alternative measures of debt sustainability are below their indicative thresholds suggesting that liquidity risks remain limited.

4. The external PPG debt outlook remains vulnerable to large shocks, in particular to a further depreciation of the exchange rate. The PV of the debt to GDP and remittances ratio raises above the relevant indicative thresholds over the medium term under all stress test scenarios (see Table AII.2). In particular, a one-time 30 percent depreciation of the som vis-à-vis the U.S. dollar, which is not unprecedented given the magnitude of the 2015 depreciation, would significantly elevate the debt ratio and result in a high risk of debt distress rating for the next five years.

Text Table AII.3. Kyrgyz Republic: Comparison of Debt Ratios

	2015	2016	2017	2018	2019	2020	Long Term (2025)
PPGE debt to GDP ratio							
Current DSA	64.6	70.2	69.6	67.0	64.8	61.4	47.5
Previous DSA (First Review)	63.0	65.3	66.1	64.7	63.6	62.7	50.8
Public debt to GDP ratio							
Current DSA	66.5	72.0	71.2	68.5	66.1	62.6	57.7
Previous DSA (First Review)	64.9	66.9	67.6	66.1	64.9	63.8	64.6

Sources: Authorities data and IMF staff calculations.

Public DSA

5. The public debt outlook has also deteriorated compared to the previous DSA, mainly driven by the increase in external public debt following the depreciation of the som. Public debt reached 66.5 percent of GDP in 2015 and is expected to increase further to above 71 percent in 2016–17 as a result of planned disbursements for public infrastructure projects. Total public debt is

[4] A borderline high/moderate case is one where stress tests result in one or more breaches, and there is a small breach (i.e., within the band) of the threshold in the baseline scenario. In the case of the Kyrgyz Republic this requires that the ratio of present value of debt to GDP and remittances falls between 36 and 37.8 percent. A borderline moderate/high case is where one or more stress tests indicate a breach, and the threshold is nearly breached in the baseline scenario (i.e., the ratio is between 34.2 and 36 percent).

[5] Under the probability approach the indicative threshold is based on country specific CPIA and GDP growth information.

expected to be manageable in the medium and long term, but remains highly sensitive to shocks that reduce real GDP growth or a failure to reduce the primary deficit over the medium term.

Conclusion

6. The 2016 joint Bank-Fund debt sustainability analysis (DSA) suggests that debt sustainability in the Kyrgyz Republic has further deteriorated compared to the last November DSA.[6] The Kyrgyz Republic is now a high-moderate borderline case, very close to its debt being rated at high risk of debt distress. A DSA assessment of high risk of debt distress would automatically reduce the availability of both concessional and nonconcessional resources,[7] and thus should be avoided. Given the persistently weak external environment and high exchange rate volatility, the borrowing space has also become extremely limited.

7. Against this background, the authorities need to remain extremely cautious when contracting and guaranteeing new debt and should resume fiscal consolidation this year without any delays. In 2016, the primary fiscal deficit is expected to be some 5 percentage points higher than the debt-stabilizing level, implying contributing to a further 5.5 percentage points increase in the public debt ratio by end 2016. In this context, it is important that the new debt limit legislation, which would allow revisions to the debt limit on an annual basis, would not be used to soften up borrowing constrains. Further efforts are needed to strengthen public debt management, in order to ensure that potential gains from externally financed public investment projects are fully utilized.

8. The authorities broadly concurred with the results of this DSA. They stressed, however, that the recent rapid debt accumulation was mainly driven by exchange rate volatility, a factor beyond their control. They argued that large scale external borrowing was necessary to the country's economic development and considered that the current debt level was sustainable to the extent that public investment projects were successful in increasing the economy's debt servicing capacity. The authorities remained committed to refrain from nonconcessional borrowing and agreed to remain prudent when contracting concessional debt.

[6] Country Report No 15/131.

[7] If assessed to be at high risk of debt distress, the current IT on zero nonconcessional borrowing would be elevated to a PC and the PC on new external debt on PV terms would be quantified in nominal terms. Financing by the WB and ADB would be restricted to grant financing and based on the "modified volume approach" a 20 percent upfront volume discount would be applied on the overall grant allocation.

Table AII.1. Kyrgyz Republic: External Debt Sustainability Framework, Baseline Scenario, 2016–36[1/]

(In percent of GDP; unless otherwise indicated)

	Actual			Historical Average [6/]	Standard Deviation [6/]	Projections									
	2013	2014	2015			2016	2017	2018	2019	2020	2021	2016-2021 Average	2026	2036	2022-2036 Average
External debt (nominal) 1/	72.9	80.5	99.0			109.1	104.6	98.8	94.5	89.4	83.9		65.7	45.2	
of which: public and publicly guaranteed (PPG)	43.7	50.6	64.6			70.2	69.6	67.0	64.8	61.4	57.2		44.8	32.2	
Change in external debt	-7.3	7.6	18.5			10.1	-4.5	-5.8	-4.3	-5.1	-5.5		-3.9	-1.3	
Identified net debt-creating flows	-15.4	11.9	9.7			3.4	5.0	0.4	0.9	-1.7	-0.5		-0.1	-1.4	
Non-interest current account deficit	0.2	16.8	9.4	4.4	6.8	12.6	13.8	11.9	11.1	8.0	7.7		7.1	4.1	6.2
Deficit in balance of goods and services	26.5	43.8	30.6			37.8	38.6	35.4	31.7	27.6	27.4		24.2	17.4	
Exports	57.7	43.5	39.8			43.2	41.6	42.3	43.2	44.5	44.8		50.9	62.6	
Imports	84.2	87.4	70.4			81.0	80.2	77.7	74.9	72.1	72.2		75.1	80.0	
Net current transfers (negative = inflow)	-31.1	-30.2	-24.3	-28.5	2.4	-27.4	-26.7	-25.2	-24.1	-22.3	-21.9		-19.2	-14.3	-17.6
of which: official	-1.7	-2.8	-1.6			-1.2	-1.4	-1.0	-0.6	0.0	0.0		0.0	0.0	
Other current account flows (negative = net inflow)	4.8	3.1	3.1			2.2	1.9	1.8	3.5	2.8	2.2		2.1	0.9	
Net FDI (negative = inflow)	-8.5	-4.7	-10.6	-7.2	2.6	-8.7	-7.3	-7.1	-7.1	-6.8	-6.6		-5.4	-4.9	-5.5
Endogenous debt dynamics 2/	-7.0	-0.2	10.9			-0.5	-1.5	-4.3	-3.1	-2.9	-1.7		-1.8	-0.6	
Contribution from nominal interest rate	1.0	1.1	1.1			2.9	1.5	0.7	1.2	2.1	2.1		2.0	1.6	
Contribution from real GDP growth	-7.6	-2.6	-3.1			-3.4	-3.0	-5.1	-4.2	-5.0	-3.8		-3.7	-2.2	
Contribution from price and exchange rate changes	-0.4	1.3	13.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	8.0	-4.3	8.8			6.7	-9.5	-6.2	-5.2	-3.4	-4.9		-3.8	0.1	
of which: exceptional financing	0.0	0.0	0.0			-0.5	-0.5	-0.4	-0.4	-0.4	-0.4		0.0	0.0	
PV of external debt 4/	...	...	75.7			84.4	80.6	76.2	73.0	69.4	65.7		51.8	34.7	
In percent of exports	...	...	190.1			195.6	193.7	180.1	169.0	155.9	146.8		101.8	55.4	
PV of PPG external debt	...	...	41.4			45.6	45.6	44.4	43.3	41.4	39.1		30.9	21.7	
In percent of exports	...	...	103.9			105.7	109.6	105.0	100.3	93.1	87.2		60.7	34.6	
In percent of government revenues	...	...	121.6			141.0	142.3	137.9	129.6	122.6	113.8		92.1	68.7	
Debt service-to-exports ratio (in percent)	17.7	15.7	16.9			24.5	21.0	17.5	17.4	18.7	18.6		16.4	9.2	
PPG debt service-to-exports ratio (in percent)	6.8	3.6	4.9			5.9	6.1	6.4	6.6	6.4	6.5		6.6	4.5	
PPG debt service-to-revenue ratio (in percent)	12.3	4.9	5.8			7.9	7.9	8.4	8.5	8.4	8.5		10.0	9.0	
Total gross financing need (Billions of U.S. dollars)	0.2	1.4	0.4			0.9	1.0	0.8	0.9	0.8	0.8		1.2	1.2	
Non-interest current account deficit that stabilizes debt ratio	7.5	9.2	-9.1			2.5	18.4	17.7	15.4	13.1	13.2		11.0	5.4	
Key macroeconomic assumptions															
Real GDP growth (in percent)	10.5	3.6	3.5	4.4	3.7	3.0	3.0	5.2	4.6	5.7	4.5	4.3	5.8	5.1	5.0
GDP deflator in US dollar terms (change in percent)	0.4	-1.7	-13.9	6.7	14.0	-14.2	4.9	3.2	2.6	2.0	2.0	0.1	2.0	2.0	2.0
Effective interest rate (percent) 5/	1.3	1.5	1.2	1.1	0.2	2.6	1.4	0.7	1.3	2.4	2.5	1.8	3.0	3.7	3.3
Growth of exports of G&S (US dollar terms, in percent)	10.3	-23.2	-18.5	12.2	24.6	-4.1	4.2	10.4	9.6	11.0	7.2	6.4	12.1	8.7	9.5
Growth of imports of G&S (US dollar terms, in percent)	12.6	5.7	-28.2	16.7	30.0	1.7	7.0	5.1	3.5	3.8	6.7	4.7	8.5	7.9	7.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	35.5	37.2	37.3	36.7	47.6	43.9	39.7	36.5	23.9	32.0
Government revenues (excluding grants, in percent of GDP)	32.0	32.1	34.0			32.4	32.1	32.2	33.4	33.8	34.3		33.6	31.5	32.9
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.4			0.4	0.4	0.4	0.3	0.3	0.2		0.4	0.5	
of which: Grants	0.2	0.2	0.3			0.3	0.3	0.3	0.2	0.2	0.2		0.2	0.3	
of which: Concessional loans	0.1	0.1	0.1			0.1	0.1	0.1	0.1	0.1	0.1		0.2	0.3	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			8.8	7.5	5.7	4.4	3.4	2.7		2.8	2.0	2.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			59.6	63.9	65.9	62.5	71.0	77.2		59.6	43.1	53.7
Memorandum items:															
Nominal GDP (Billions of US dollars)	7.3	7.5	6.7			5.9	6.4	6.9	7.4	8.0	8.5		11.9	23.6	
Nominal dollar GDP growth	11.0	1.8	-10.9			-11.6	8.0	8.6	7.4	7.8	6.6	4.5	7.9	7.1	7.0
PV of PPG external debt (in Billions of US dollars)			2.31			2.6	2.9	3.0	3.2	3.3	3.3		3.6	5.1	
(PVt-PVt-1)/GDPt-1 (in percent)						4.9	4.1	2.6	2.1	1.3	0.2	2.5	0.5	1.0	0.8
Gross workers' remittances (Billions of US dollars)	2.2	2.0	1.5			1.5	1.6	1.7	1.7	1.8	1.9		2.3	3.4	
PV of PPG external debt (in percent of GDP + remittances)	...	...	33.7			36.1	36.4	35.8	35.1	33.9	32.0		25.9	18.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	66.1			65.7	68.2	66.8	65.0	62.0	58.6		44.1	28.2	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	3.1			3.7	3.8	4.1	4.3	4.3	4.4		4.8	3.7	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

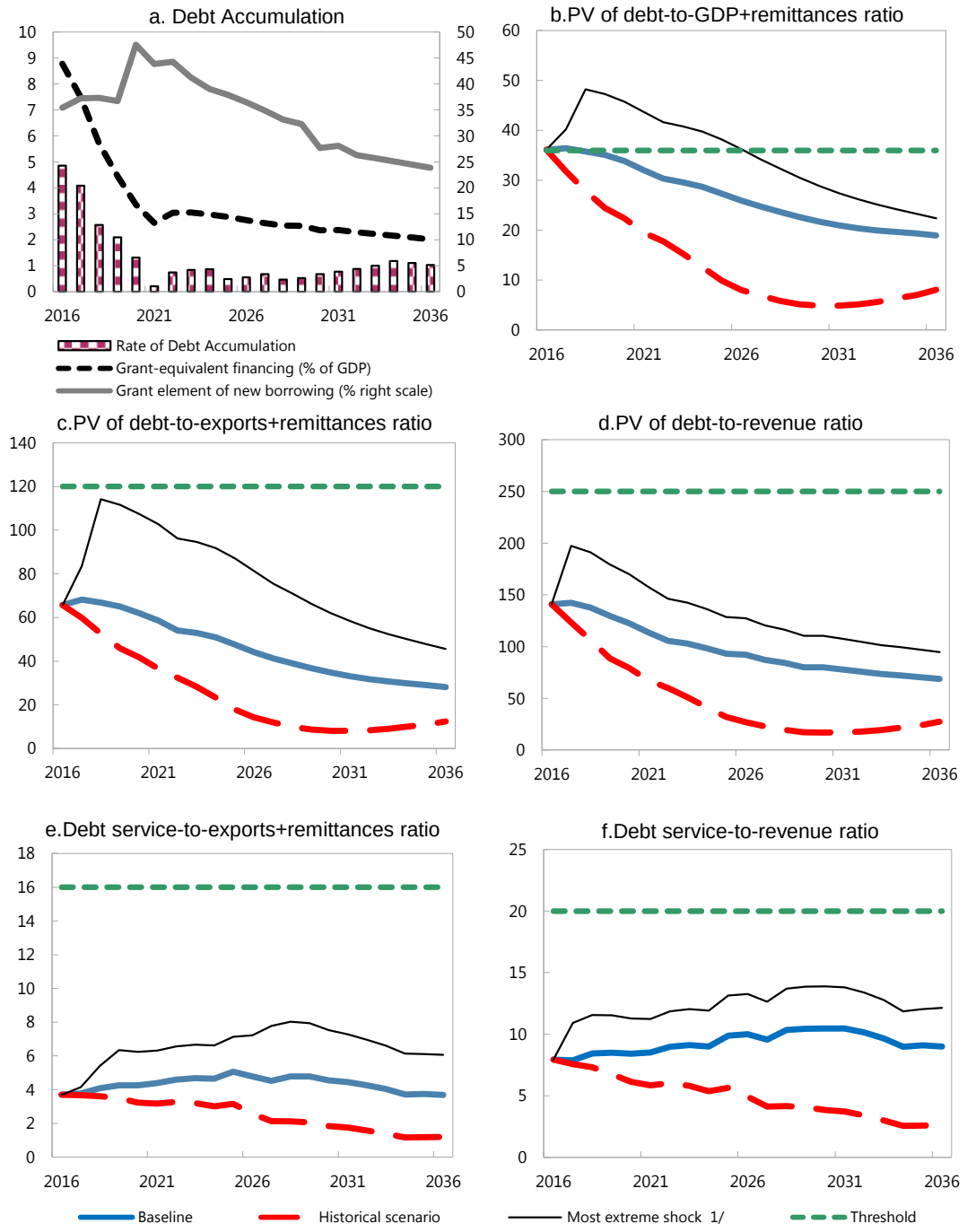
5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

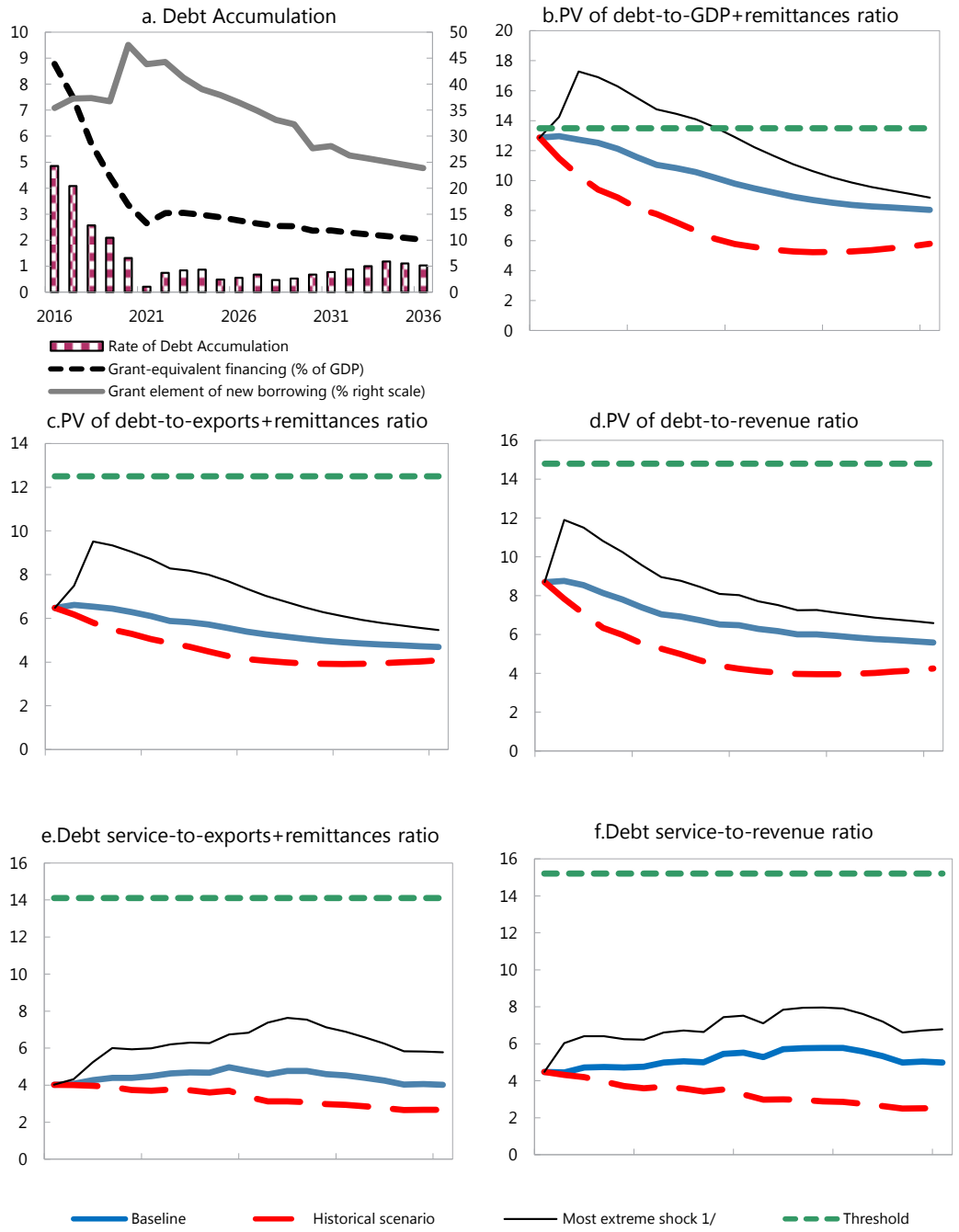
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Figure AII.1. Kyrgyz Republic: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016–36 /1



Sources: Country authorities; and staff estimates and projections.

Figure AII.2. Kyrgyz Republic: Probability of Debt Distress of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2016–36 /1



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Table AII.2. Kyrgyz Republic: Indicators of Public and Publicly-Guaranteed External Debt Under Alternative Scenarios, 2016–36[1/]

	Projections							
	2016	2017	2018	2019	2020	2021	**2026**	2036
PV of debt-to-GDP+remittances ratio								
Baseline	36.1	36.4	35.8	35.1	33.9	32.1	**26.2**	19
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	36	32	28	24	22	20	**8**	8
A2. New public sector loans on less favorable terms in 2016-2036 2	36	37	37	38	37	35	**32**	32
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	36	37	37	37	35	34	**27**	20
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	36	40	48	47	46	44	**37**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	36	40	42	42	40	38	**31**	23
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	36	38	38	38	36	35	**28**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	36	40	47	45	44	42	**35**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	36	47	46	45	44	42	**34**	25
PV of debt-to-exports+remittances ratio								
Baseline	66	68	67	65	62	59	**45**	28
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	66	60	53	46	42	37	**14**	13
A2. New public sector loans on less favorable terms in 2016-2036 2	66	70	70	70	67	64	**54**	47
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	66	68	66	64	61	58	**44**	28
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	66	84	114	112	108	103	**82**	45
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	66	68	66	64	61	58	**44**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	66	72	71	70	67	63	**48**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	66	74	85	82	79	75	**59**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	66	68	66	64	61	58	**44**	28
PV of debt-to-revenue ratio								
Baseline	141	142	138	130	123	114	**93**	69
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	141	123	106	89	79	67	**28**	27
A2. New public sector loans on less favorable terms in 2016-2036 2	141	146	144	139	133	125	**113**	115
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	141	144	146	137	130	121	**98**	73
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	141	157	186	175	166	156	**130**	80
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	141	160	172	162	153	142	**116**	86
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	141	149	148	139	132	123	**101**	71
B5. Combination of B1-B4 using one-half standard deviation shocks	141	160	187	176	167	156	**129**	85
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	141	197	191	180	170	158	**129**	95

Table AII.2. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public and Publicly-Guaranteed External Debt, 2016–36 (continued)[1/]

Baseline	4	4	4	4	4	4	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	4	4	4	3	3	3	**3**	1
A2. New public sector loans on less favorable terms in 2016-2036 2	4	4	4	4	4	4	**5**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	4	4	4	4	4	4	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	4	4	5	6	6	6	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	4	4	4	4	4	4	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	4	4	4	4	4	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	4	5	5	5	**5**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	4	4	4	4	4	4	**5**	4
Debt service-to-revenue ratio								
Baseline	8	8	8	8	8	9	**10**	9
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2016-2036 1/	8	8	7	7	6	6	**5**	3
A2. New public sector loans on less favorable terms in 2016-2036 2	8	8	8	8	8	9	**10**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2017-2018	8	8	9	9	9	9	**10**	9
B2. Export value growth at historical average minus one standard deviation in 2017-2018 3/	8	8	9	10	10	10	**11**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2017-2018	8	9	10	10	10	10	**12**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2017-2018 4/	8	8	9	9	8	8	**10**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	10	10	10	10	**12**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2017 5/	8	11	12	12	11	11	**13**	12
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	22	22	22	22	22	22	**22**	22

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table AII.3. Kyrgyz Republic: Public Sector Debt Sustainability Framework, Baseline Scenario, 2013–36

(In percent of GDP; unless otherwise indicated)

	Actual					Estimate	Projections								
	2013	2014	2015	Average 5/	Standard Deviation 5/	2016	2017	2018	2019	2020	2021	2016-21 Average	2026	2036	2022-36 Average
Public sector debt 1/	46.1	52.6	66.5			72.0	71.2	68.5	66.1	62.6	58.3		56.1	50.2	
of which: foreign-currency denominated	43.7	50.6	64.6			70.2	69.6	67.0	64.8	61.4	57.2		44.8	32.2	
Change in public sector debt	-2.9	6.5	13.9			5.5	-0.9	-2.7	-2.4	-3.5	-4.2		-1.6	-2.9	
Identified debt-creating flows	-1.0	6.2	14.3			6.0	-1.3	-3.3	-3.1	-4.6	-3.8		-1.9	-1.4	
Primary deficit	2.7	2.8	2.3	2.8	2.0	7.4	4.1	1.7	1.0	-0.5	-0.6	2.2	1.2	1.0	1.2
Revenue and grants	34.4	35.3	38.1			37.9	37.0	36.2	36.3	35.9	36.4		35.2	32.7	
of which: grants	2.4	3.3	4.1			5.5	5.0	4.0	2.9	2.1	2.0		1.7	1.2	
Primary (noninterest) expenditure	37.2	38.2	40.4			45.3	41.2	37.9	37.3	35.4	35.7		36.5	33.7	
Automatic debt dynamics	-3.8	3.3	12.0			-0.9	-4.9	-4.6	-3.7	-3.7	-2.8		-3.1	-2.4	
Contribution from interest rate/growth differential	-5.2	-2.0	-1.7			-1.9	-2.4	-3.8	-3.2	-3.7	-2.8		-3.1	-1.7	
of which: contribution from average real interest rate	-0.5	-0.4	0.1			0.1	-0.3	-0.2	-0.2	-0.1	-0.1		0.1	0.8	
of which: contribution from real GDP growth	-4.7	-1.6	-1.8			-2.0	-2.1	-3.5	-3.0	-3.6	-2.7		-3.2	-2.6	
Contribution from real exchange rate depreciation	1.4	5.3	13.7			1.0	-2.6	-0.8	-0.5	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			-0.5	-0.5	-0.4	-0.4	-0.4	-0.4		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			-0.5	-0.5	-0.4	-0.4	-0.4	-0.4		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.8	0.3	-0.4			-0.5	0.4	0.6	0.7	1.2	-0.4		0.3	-1.5	
Other Sustainability Indicators															
PV of public sector debt	...	2.0	43.3			47.4	47.2	45.9	44.6	42.6	40.2		42.2	39.6	
of which: foreign-currency denominated	...	0.0	41.4			45.6	45.6	44.4	43.3	41.4	39.1		30.9	21.7	
of which: external	...	...	41.4			45.6	45.6	44.4	43.3	41.4	39.1		30.9	21.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	7.7	6.1	5.5			11.3	7.6	5.3	4.6	3.0	2.8		10.4	2.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	5.7	113.7			125.1	127.5	126.8	122.9	118.7	110.4		119.7	121.2	
PV of public sector debt-to-revenue ratio (in percent)	...	6.3	127.2			146.5	147.3	142.3	133.5	126.2	117.0		125.7	125.7	
of which: external 3/	...	...	121.6			141.0	142.3	137.9	129.6	122.6	113.8		92.1	68.7	
Debt service-to-revenue and grants ratio (in percent) 4/	11.6	5.7	5.9			7.9	7.2	7.8	8.1	8.3	8.2		15.6	12.1	
Debt service-to-revenue ratio (in percent) 4/	12.5	6.3	6.7			9.2	8.4	8.8	8.8	8.8	8.7		16.4	12.6	
Primary deficit that stabilizes the debt-to-GDP ratio	5.6	-3.7	-11.6			1.9	5.0	4.4	3.4	3.0	3.6		2.8	3.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	10.5	3.6	3.5	4.4	3.7	3.0	3.0	5.2	4.6	5.7	4.5	4.3	5.8	5.1	5.0
Average nominal interest rate on forex debt (in percent)	1.2	1.3	1.3	1.1	0.2	1.5	1.4	1.5	1.5	1.6	1.7	1.5	2.1	2.6	2.3
Average real interest rate on domestic debt (in percent)	16.7	5.1	13.3	2.9	8.8	18.3	1.9	1.9	1.9	1.9	1.9	4.6	0.0	0.0	0.0
Real exchange rate depreciation (in percent, + indicates depreciation)	3.4	12.8	28.1	-0.7	14.6	1.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.5	8.9	2.2	10.8	7.4	4.3	8.1	5.3	4.7	4.0	4.0	5.1	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	4.6	6.3	9.4	2.1	3.4	15.6	-6.4	-3.1	3.1	0.1	5.6	2.5	0.8	5.2	4.6
Grant element of new external borrowing (in percent)	...	...	0.0	0.0	...	35.5	37.2	37.3	36.7	47.6	43.9	39.7	36.5	23.9	...

Sources: Country authorities; and staff estimates and projections.

1/ Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.

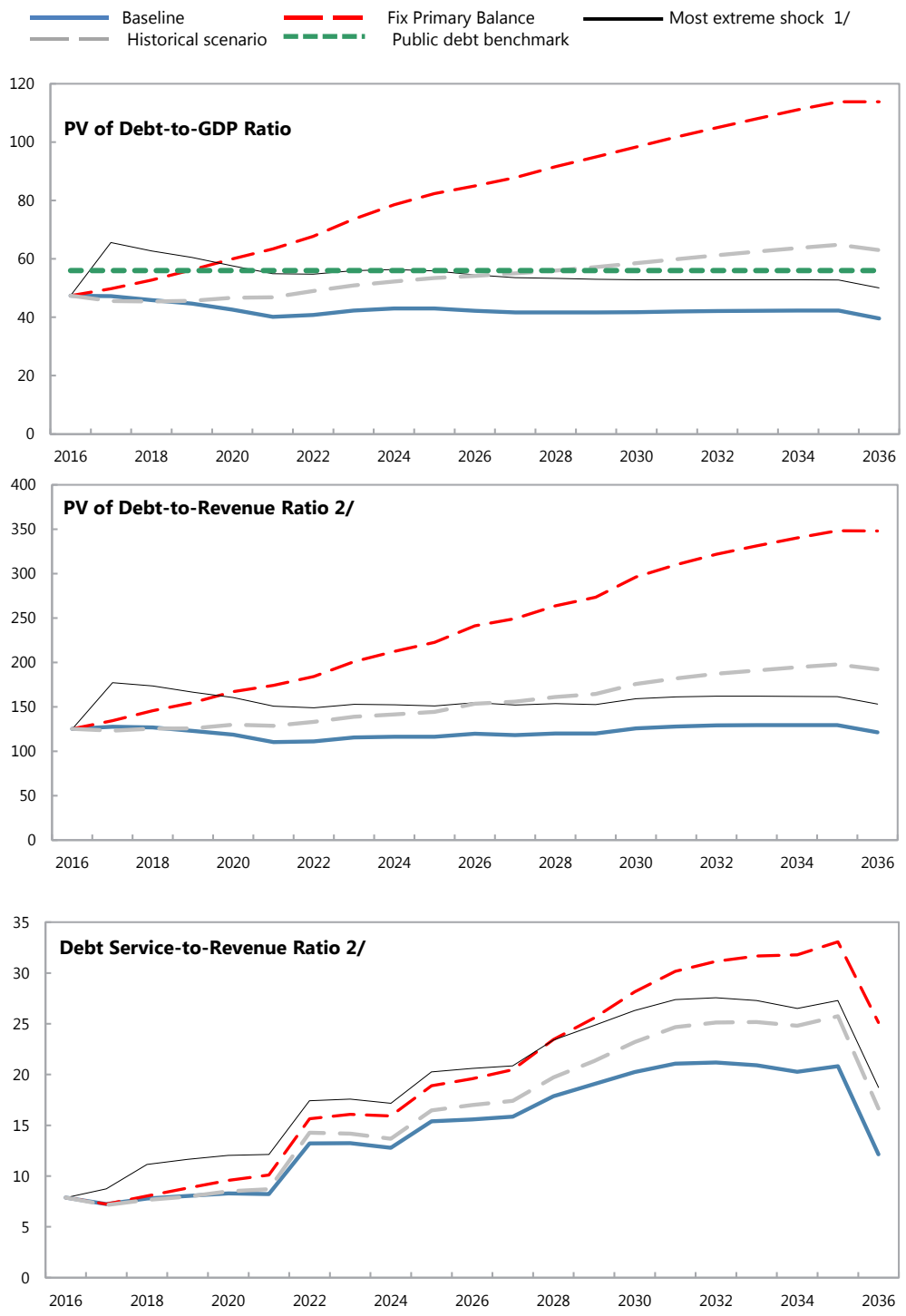
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Figure AII.3. Kyrgyz Republic: Indicators of Public Debt Under Alternative Scenarios, 2016–36 /1



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2026.

2/ Revenues are defined inclusive of grants.

Table AII.4. Kyrgyz Republic: Sensitivity Analysis for Key Indicators of Public Debt, 2016–36

	Projections							
	2016	2017	2018	2019	2020	2021	2026	2036
PV of Debt-to-GDP Ratio								
Baseline	47	47	46	45	43	40	42	40
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	46	45	46	47	47	54	63
A2. Primary balance is unchanged from 2016	47	50	53	56	60	63	85	114
A3. Permanently lower GDP growth 1/	47	48	47	47	46	45	57	84
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	47	49	51	52	51	50	60	68
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	47	48	49	47	45	43	45	41
B3. Combination of B1-B2 using one half standard deviation shocks	47	47	48	48	47	45	50	52
B4. One-time 30 percent real depreciation in 2017	47	66	63	60	58	55	55	50
B5. 10 percent of GDP increase in other debt-creating flows in 2017	47	55	53	52	50	47	48	44
PV of Debt-to-Revenue Ratio 2/								
Baseline	125	127	127	123	119	110	120	121
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	125	123	126	126	130	129	153	192
A2. Primary balance is unchanged from 2016	125	134	146	155	167	174	241	348
A3. Permanently lower GDP growth 1/	125	129	130	129	128	123	160	256
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	125	131	141	141	142	138	170	208
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	125	129	135	131	126	118	126	126
B3. Combination of B1-B2 using one half standard deviation shocks	125	127	134	132	130	123	143	160
B4. One-time 30 percent real depreciation in 2017	125	177	173	166	160	151	155	153
B5. 10 percent of GDP increase in other debt-creating flows in 2017	125	148	147	143	138	129	137	134
Debt Service-to-Revenue Ratio 2/								
Baseline	8	7	8	8	8	8	16	12
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	7	8	8	8	9	17	17
A2. Primary balance is unchanged from 2016	8	7	8	9	10	10	20	25
A3. Permanently lower GDP growth 1/	8	7	8	8	9	9	18	19
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2017-2018	8	7	8	9	9	9	18	17
B2. Primary balance is at historical average minus one standard deviations in 2017-2018	8	7	8	8	9	8	16	13
B3. Combination of B1-B2 using one half standard deviation shocks	8	7	8	8	9	9	17	15
B4. One-time 30 percent real depreciation in 2017	8	9	11	12	12	12	21	19
B5. 10 percent of GDP increase in other debt-creating flows in 2017	8	7	9	9	9	9	16	13

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.